

Bitcoin Market Sentiment vs Trader Performance

An in-depth analysis of how Bitcoin's Fear & Greed Index correlates with trader profitability and behavior – using Hyperliquid historical trading data.

PREPARED BY: GOURAV MALVIYA



Objective

What We're Investigating

This project analyzes the relationship between **Bitcoin market sentiment** – measured by the Fear & Greed Index – and **trader performance** using Hyperliquid historical trading data. The goal is to identify patterns that can help improve trading strategies.

Core Question

How does trader profitability change across different market sentiment conditions? By examining trading behavior, activity patterns, and performance across coins and trading sessions, this analysis aims to surface insights that support **data-driven trading and risk management decisions**.

Datasets Used

Two complementary datasets power this analysis, combining market sentiment signals with granular trader activity.



Bitcoin Fear & Greed Index Dataset

A sentiment indicator that quantifies market emotion on a scale from extreme fear to extreme greed. Used to classify trading conditions and segment performance by sentiment regime.



Hyperliquid Historical Trader Dataset

Historical trading records from the Hyperliquid exchange, capturing individual trader P/L, buy/sell activity, coin exposure, and timestamp data for behavioral and performance analysis.

Methodology

A structured analytical pipeline was applied to ensure clean, reliable, and insightful results. Each stage builds on the previous to transform raw data into actionable findings.



Data Cleaning

Remove inconsistencies, handle missing values, and standardize formats across both datasets.



Date Conversion

Align timestamps across datasets to enable accurate merging and time-based analysis.



Dataset Merging

Join Fear & Greed Index data with trader records on matching dates for sentiment-labeled analysis.



Feature Engineering

Create derived variables such as sentiment categories, session labels, and aggregated metrics.



EDA & Testing

Exploratory data analysis, statistical testing, and business insight generation from results.

Key Analyses — Performance & Behavior

Profitability Analysis

Profit/Loss Distribution

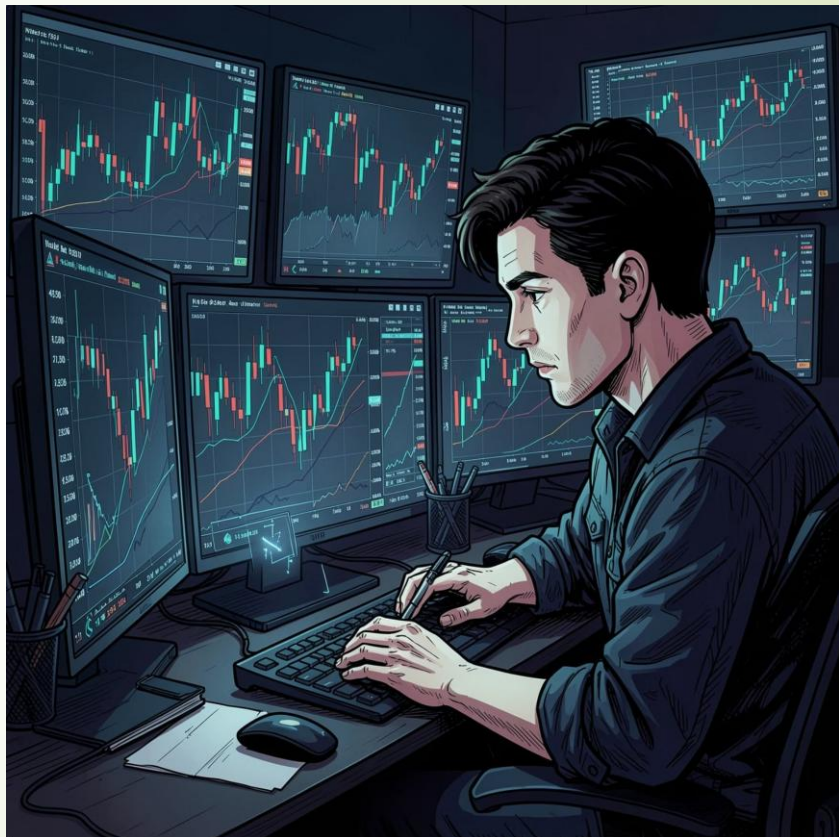
Examining the spread of trader P/L to understand overall performance concentration and outlier behavior.

Profit by Market Sentiment

Segmenting trader returns across Fear, Neutral, and Greed regimes to detect sentiment-driven performance patterns.

Buy vs Sell Comparison

Comparing profitability of long (buy) versus short (sell) positions to identify directional biases in trader success.



Key Analyses — Activity & Correlation

Trading Activity by Hour

Identifying peak and off-peak trading hours to understand when traders are most active and whether timing correlates with performance.

Trading Activity by Weekday

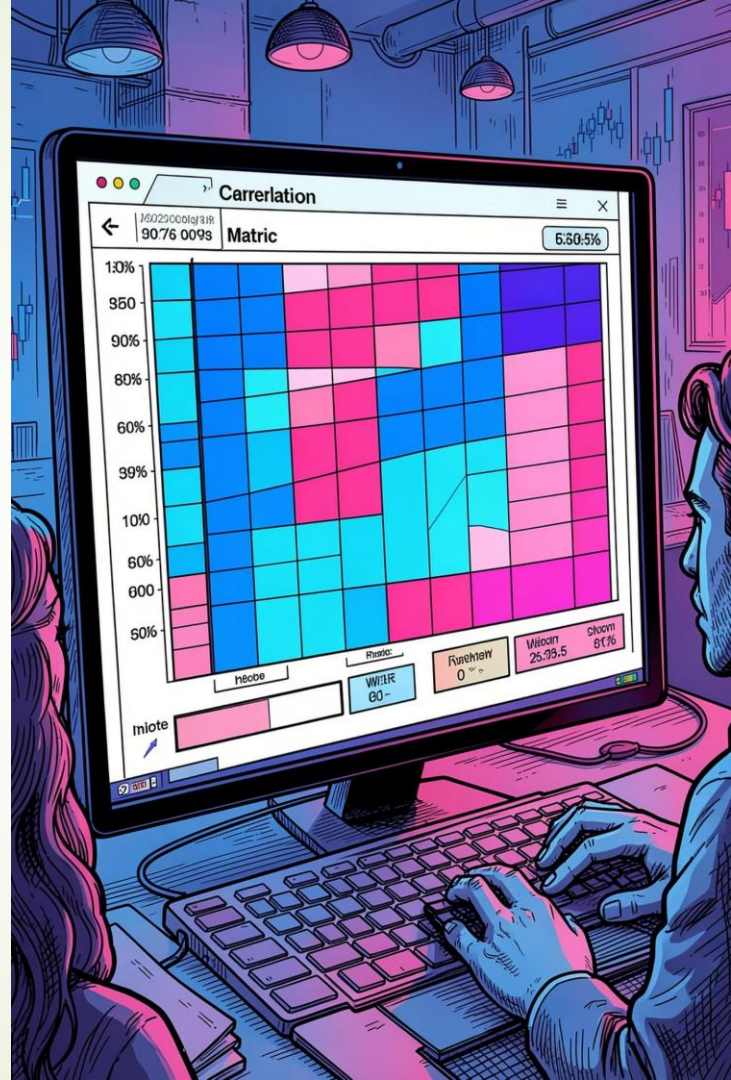
Analyzing volume and P/L patterns across days of the week to detect weekly seasonality in trader behavior.

Top Traders & Top Coins

Ranking traders by profitability and coins by trading volume to highlight standout performers and most-traded assets.

Correlation & Statistical Testing

Measuring relationships between sentiment scores and trader metrics, with Fear vs Greed statistical comparison to validate observed patterns.





Key Insights

This analysis investigates how trader profitability changes across different market sentiment conditions, exploring trading behavior, activity patterns, and performance across coins and trading sessions. The notebook examines:

1 Sentiment-Driven Profitability

Whether trader returns differ significantly under Fear, Neutral, and Greed market conditions – and which sentiment regime favors certain strategies.

2 Behavioral Patterns

How trading activity varies by hour and weekday, revealing when traders are most active and whether timing aligns with better outcomes.

3 Coin & Trader Performance

Which coins attract the most trading volume and which traders demonstrate consistent profitability across sentiment conditions.

Final conclusions should be based on the results produced by the notebook.

Conclusion

What This Project Delivers

A structured, reproducible analysis of trader performance under varying Bitcoin market sentiment conditions – combining sentiment data with real trading records to surface evidence-based patterns.

Practical Applications

The findings can support **data-driven trading decisions** by revealing how sentiment regimes affect profitability, and **risk management strategies** by highlighting periods of elevated risk or opportunity. Traders and analysts can use these insights to refine entry/exit timing, position sizing, and coin selection based on prevailing market emotion.

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Datasets Combined

Fear & Greed Index + Hyperliquid trader records

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Analysis Types

From P/L distribution to correlation testing

1

Goal

Data-driven trading and risk management decisions

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